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Temperature Predicts Relative Mass but Not Size in Passerines

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ABSTRACT

Aim: Predictable relationships between temperature and animal size and shape are the basis for classic rules in macroecology. Body size is thought to covary negatively with temperature (Bergmann's Rule), broadly attributed to thermoregulatory demand for larger size in colder climates. However, ecophysiological principles suggest that changes in insulative materials may be a more effective thermoregulatory adaptation than changes in overall skeletal size. We test for global gradients in mass relative to skeletal size (a proxy for insulation) and body size.

Location: Global.

Time Period: Current.

Major Taxa Studied: Aves: Passeriformes.

Methods: Combining a dataset of mass and skeletal traits for 1958 passerine species, we use Bayesian phylogenetic comparative analyses to test the extent to which temperature predicts skeletal size and relative mass.

Results: Across species, we find no evidence that temperature predicts skeletal size; rather, the effects associated with phylogeny account for nearly all of the variation in size. In contrast, we find a global gradient in mass relative to skeletal size, with species in colder climates having more mass than expected given their skeletal size. For 24 species, we assess these relationships within species and find that relative mass has declined through time but is not predicted by temperature.

Main Conclusions: Mass relative to skeletal size is expected to exert a significant effect on insulation and energy stores, and we find it is more labile than skeletal size across species. Our results suggest: (1) species in colder climates tend to have greater insulation (as indicated by their higher relative mass); (2) skeletal size in passerines does not strongly adhere to Bergmann's Rule, suggesting spatial temperature-size relationships may be unlikely to predict temporal responses of size to climate change in this group; and (3) declines in relative mass may be a widespread cryptic response to global change.

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1 | Introduction

Animal phenotypes are the outcomes of the interplay between evolutionary and ecological processes operating across a range of temporal scales. Among these, climate has long been viewed as influential in determining morphology. For over a century, ecologists have explored Bergmann's Rule (Bergmann 1847): the tendency for endotherms to be larger in colder climates, broadly attributed to increased heat conservation efficiency resulting from a smaller surface area to volume ratio in larger individuals (Bergmann 1847; Watt and Salewski 2011). Given the primacy of size as an ecological trait (Brown et al. 2017; McGill et al. 2006) and the increasing desire to use spatial relationships between temperature and size to predict morphological responses to climate change (Tian and Benton 2020), understanding the generality of temperature-size relationships and their underlying causes and implications is critical.

Despite its long-standing influence, the empirical support for Bergmann's Rule is mixed. The consistency with which species are found to follow Bergmann's Rule varies widely, with some meta-analyses putting the number as high as 75% in mammals and 90% in birds (Ashton 2002; Ashton et al. 2000), while other studies looking broadly across endotherms have found as few as 14% of species follow Bergmann's Rule (Riener et al. 2018). Efforts to move beyond estimating the proportion of species that adhere to Bergmann's Rule to understand the mechanistic basis for geographical clines have been comparatively limited (Tattersall et al. 2012). Moreover, an increasing number of recent comparative assessments that disentangle the predictive power of climate variables and phylogenetic history (e.g., He et al. 2023; Mainwaring and Street 2021) have demonstrated that history accounts for orders of magnitude more variation in size than climatic variables (e.g., 0.04% variation in size explained by climate compared to 96% explained by history; Mainwaring and Street 2021). This mixed evidence of the generality of the rule along with limited direct empirical data on the causal basis for temperature-size relationships raises important questions about the role of temperature in shaping size variation.

A parallel approach to understanding the influence of temperature on size evolution has been to confront the question from an ecophysiological perspective. Physiologists have long questioned the thermoregulatory relevance of shifts in surface area to volume ratios that would result from observed size variation within closely related individuals (McNab 1971; Nord et al. 2024; Scholander 1955). Allometric scaling laws require substantial increases in body size for even small decreases in heat flux (Geist 1987; Nord et al. 2024). Since the magnitude of geographical size clines can be quite small (He et al. 2023; Henry et al. 2023; Mainwaring and Street 2021), the thermoregulatory benefit of observed body size variation with temperature may be minimal (Tabh et al. 2025).

While subtle size variation is expected to have a relatively small impact on thermoregulation, increases in insulative layers (such as fur, feathers, or fat) consistently appear to produce large thermoregulatory effects (Steudel et al. 1994). In some species, the same heat savings gained by doubling body size could be attained by increasing fur depth by only 4 mm (Briscoe

et al. 2015). Further, insulation tends to be highly dynamic compared to skeletal size (Broggi et al. 2011; Nord et al. 2023; Scott et al. 1994). Feathers and fur are shed and replaced throughout the year (Beltran et al. 2018), with winter pelages and plumages often having increased insulative capacity (Nord et al. 2023; Zimova et al. 2018), and body fat can change rapidly over the course of a year or day (Blem 1976; Scott et al. 1994). Moreover, phylogenetic signal in insulative traits tends to be moderate to low (Barve et al. 2021; Fristoe et al. 2015; Heldstab et al. 2017; Riddell et al. 2022), suggesting that species-level insulative traits may readily adapt to species' thermal conditions in addition to individual variation across the lifespan or year (Riddell et al. 2022). In contrast, skeletal size in endotherms tends to exhibit strong phylogenetic conservatism (Morales-Castilla et al. 2012), and mostly does not change after adult size is attained (Sebens 1987). Thus, given the increased effectiveness of insulation as a cold tolerance strategy and its relative flexibility, it may be an axis of adaptation that endotherms can easily move along to adjust to novel climates whereas skeletal size may be more limited.

Despite the importance of insulation to thermoregulation, it has been difficult to quantify relationships between climatic variables and variation in insulation at large spatial scales and across diverse sets of taxa. Common measurements like mass do not distinguish between body composition and skeletal size (Piersma and Davidson 1991), and innovations in measuring body composition like quantitative magnetic resonance (Guglielmo et al. 2022; McGuire and Guglielmo 2010) have yet to be deployed at sufficiently broad scales to test for geographical variation in body composition within species. Studies examining variation within single taxa (Blem 1974; Broggi et al. 2011; Koskenpato et al. 2016; Scholander et al. 1950; Swanson 1991), at regional scales (Barve et al. 2021), and across species (Fristoe et al. 2015; Osváth et al. 2018; Wiersma et al. 2012) often find geographic clines in insulative materials linked to temperature, suggesting that a broad spatial pattern may exist that has not yet been tested at a large taxonomic and spatial scale.

Birds are increasingly becoming a model system for understanding spatial and temporal relationships between climate and morphology. Within birds, roughly 6500 species are passerines (Order: Passeriformes), comprising approximately 60% of avian diversity (Schmitt and Edwards 2022). Their relatively stable taxonomy and well-known phylogenetic relationships, distributions, ecologies, and morphologies make them an excellent system for studying variation in morphological adaptations to climate. We build on a new skeletal trait dataset of 14,419 passerine skeletons (Weeks, Zhou, et al. 2025), to test associations between climate and morphology in 1958 species (approximately 30% of passerines; Figure 1). Combining this skeletal dataset with mass data (Tobias et al. 2022) allows us to calculate mass relative to skeletal size, a proxy for insulation, at an unprecedented scale within birds.

Using these data, we test the hypotheses that both within and across species, in cooler climates: (1) skeletal size will be larger (Bergmann's Rule), and (2) mass will be greater than expected given skeletal size. Furthermore, we hypothesise that temperature will explain more variation in mass relative to skeletal size

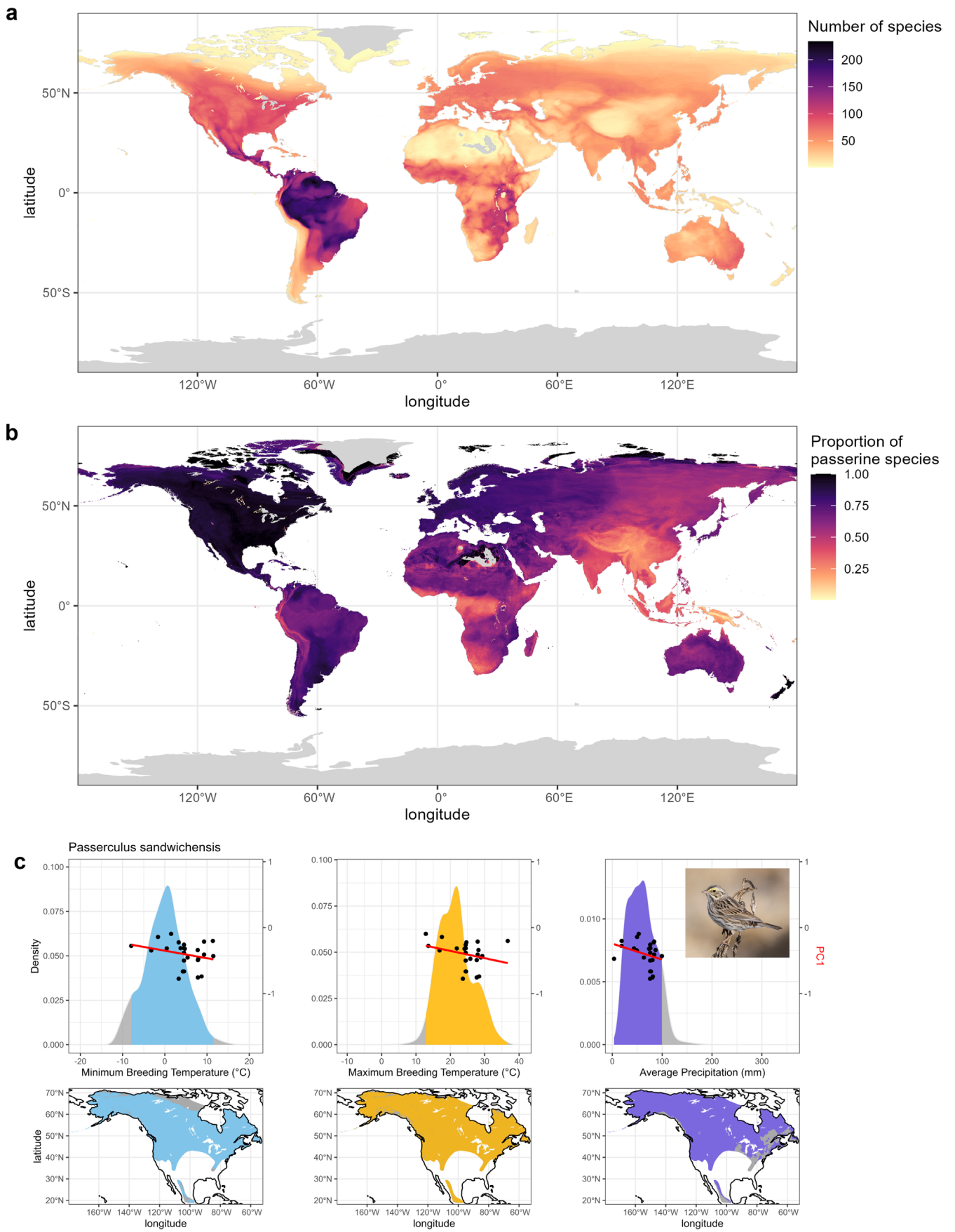


FIGURE 1 | Legend on next page.

FIGURE 1 | Geographic and climatic sampling distributions. (a) The number of species included in our dataset across the globe with darker regions representing areas with a higher number of species in our dataset, based on BirdLife range maps (BirdLife International 2015). (b) The proportion of sampled passerines versus total passerine richness, with darker regions representing areas with higher proportional representation. Passerine richness data used to generate this map drawn from BiodiversityMapping.org (Jenkins et al. 2013; Pimm et al. 2014), based on BirdLife range maps (BirdLife International 2015). Although our sample broadly corresponds to geographical passerine diversity, there is large-scale variation in the proportion of total passerine diversity sampled. (c) Intraspecific sampling for a representative species, the Savannah Sparrow (*Passerculus sandwichensis*). For the intraspecific models, sampled specimens span the breadth of the breeding range conditions. Density plots show the distribution of environmental conditions across the breeding range, with the coloured portions indicating the range of values for each variable that are covered by our sampling. Points and trendline illustrate the relationship between PC1 and each environmental variable. This species is typical of our sampling; corresponding figures for each species are in Figure S7. Savannah Sparrow image Brian Calk/Macaulay Library at the Cornell Lab of Ornithology (ML136429501).

than skeletal size itself, reflecting the differing contributions of each trait to thermoregulation as predicted by ecophysiological first principles.

2 | Materials and Methods

2.1 | Skeletal Data Collection

We estimated skeletal size using a dataset that was constructed using computer vision to measure 14,419 passerine skeletal specimens from the University of Michigan Museum of Zoology ($N=12,421$) and the Field Museum of Natural History ($N=1998$) (Weeks et al. 2023; Weeks, Zhou, et al. 2025). Additionally, to have adequate within-species sampling to fit intraspecific models, we supplemented the Weeks, Zhou, et al. (2025) dataset by measuring the same skeletal traits on 446 specimens at the American Museum of Natural History (Table S1). A single person (CMP) measured all traits using digital callipers. Our final sample comprised 1958 passerine species with a worldwide distribution (Figure 1a,b), representing 86 families and 706 genera (90% and 60% of all families and genera in passerines, respectively, Tobias et al. 2022). The number of specimens sampled within each species ranged from 1 to 120, with a median of 3.

For the within-species analyses, we only analysed those specimens that had been collected in the breeding range and for which coordinate or locality information was available. For migratory birds, we additionally only used those specimens that were collected during the breeding season to ensure that the individual bird was likely a local breeder experiencing the climate conditions at the collection location. If a locality, but not coordinate information, was available, we georeferenced the specimen using GEOLocate (www.geo-locate.org). We only included species with at least 15 unique latitudes represented. This resulted in a final sample of 57 species for the intraspecific models. The number of specimens sampled within each species ranged from 15 to 99, with a median of 28.

2.2 | Morphological Metrics

To estimate skeletal size, we took the first axis of a principal components analysis (PC1) of the lengths of the skull, keel, tarsus, and humerus. PC1 explained 92.26% of the variation, reflecting its reliability as an indicator of skeletal size, and all loadings were positive such that higher PC1 scores reflect larger

size (Rising and Somers 1989) (Table S2). For the interspecific analysis, for each species, the PC1 scores for all specimens of that species were averaged to create a species average PC1 score. We added the absolute value of the minimum PC1 score plus 1 to each score to make all scores positive, then applied a log transformation to improve residual normality. While the log transformation improved the normality of the distribution of the dependent variable and residual error, as a sensitivity analysis we additionally ran our model using the non-transformed PC1 scores, which produced qualitatively similar results (Figure S1).

To calculate mass relative to skeletal size, we followed a standard method for estimating body condition (Kraft et al. 2019). For each individual, we calculated a scaled mass residual representing the degree to which a species has more or less mass than predicted by its skeletal size. In our data, mass scales with skeletal size following a 1/3 power law (Figure S2a). We fit a linear model predicting mass from PC1, with a cube root transformation applied to mass to linearize the relationship. Allometric relationships are inherently heteroscedastic (Dutca et al. 2022). To correct for this, each residual was divided by its predicted value, resulting in the scaled mass residuals used in further analyses (Figure S2d). The scaled mass residual therefore represents the percent deviance from expected mass given the observed skeletal size; we refer to this measurement as “relative mass”. Definitionally, relative mass is not correlated with PC1; a bird with a small body size can have large relative mass and vice versa (Figure S2d).

Because not all specimens had associated mass data, we used the species mass provided by AVONET (Tobias et al. 2022) for the interspecific model. For the intraspecific model, we used individual specimen masses recorded in VertNet. Although individual mass may decrease slightly between time of collection and time of preparation due to dehydration, we do not expect that this would have varied systematically in any way that would affect our analyses. Due to the lower number of specimens with associated mass data, the sample size was smaller for the intraspecific relative mass model (24 species) than it was for the intraspecific skeletal size model (57 species).

2.3 | Climate and Life History Variable Extraction

We began by calculating maximum temperature, minimum temperature, precipitation, and normalized difference vegetation index (NDVI; a proxy for primary productivity) across

the portion of each species' range that is occupied during each month of the year. For birds with AVONET centroid breeding range latitudes north of the equator, we assumed the breeding season was May–August. For birds with centroid breeding range latitudes south of the equator, we assumed the breeding season was October–January. We acknowledge that this is a simplification, particularly as birds near the equator exhibit many different breeding strategies, including year-round breeding (Stouffer et al. 2013). To ensure that this approximates the breeding months of these species, we randomly sampled 100 species from our dataset that had centroid breeding latitudes equal to or less than zero degrees, as well as 100 species with centroid breeding latitudes between 0 and 10 degrees. We then verified their months of breeding ground occupancy in Birds of the World (Billerman et al. 2022). The majority of austral birds in our sub-sample were on the breeding grounds from October to January, reflecting the appropriateness of using this time period as the breeding season for austral breeding birds. This also aligns with a peak in breeding activity observed in the Amazon (Stouffer et al. 2013). Similarly, the majority of low-latitude northern hemisphere birds in the sub-sample were breeding May–August, suggesting this is a reasonable approximation of breeding dates for these species.

For the interspecific analysis, we used WorldClim version 2.1 1970–2000 average rasters for maximum temperature, minimum temperature, and precipitation (Fick and Hijmans 2017), at a resolution of 2.5 min. For spatial data processing, we used the R packages **raster** (Hijmans 2025), **terra** (Hijmans et al. 2026), **sf** (Pebesma 2018; Pebesma and Bivand 2023), **ncdf4** (Pierce 2025), and **fasterize** (Ross 2024). We extracted climate variables from the combined resident and breeding ranges for each species for the breeding months, and from the combined resident and non-breeding ranges for the nonbreeding months (Figure S3a–c). This was done by overlaying the relevant species range polygons (BirdLife International 2015) onto the climate rasters and calculating the average cell value (Figure S3a–c). Such an approach allows us to estimate the average temperatures experienced at different points in the year for each species. This is especially relevant for migratory birds, which may experience the coldest temperatures of their annual cycle on their breeding grounds (e.g., May in Alaska), and the warmest temperatures on their nonbreeding grounds (e.g., December in Ecuador) (Dufour et al. 2020). We defined annual maximum and minimum temperatures as the average maximum temperature during the hottest month and the average minimum temperature during the coolest month, respectively.

NDVI data from 1981 to 2000 were obtained from NOAA Climate Data Record AVHRR Normalized Difference Vegetation Index at a resolution of 0.05×0.05 degrees (Vermote and NOAA CDR Program 2019). We calculated the maximum NDVI of the breeding season (May–August or October–January) for each raster cell, then took the average of these raster cells across all years as a representation of ecologically relevant net primary productivity (Huston and Wolverton 2011) (Figure S3d). We used AVONET (Tobias et al. 2022) to determine migratory status, habitat, range size, centroid latitude, and mass for each species.

For the intraspecific analyses, we calculated the maximum temperature, minimum temperature, and average precipitation of

each collection location during the breeding season prior to collection (Figure S3e–g). We used CHESAcruts 30 arc sec resolution rasters (Karger et al. 2017; Karger and Zimmermann 2018). Because NDVI data are not available prior to 1980, we used the same long-term average maximum NDVI raster that was calculated for the interspecific analysis and extracted the NDVI value for each collection location (Figure S3h).

Process models demonstrating the calculation of all climate variables used are available in the supplement (Figure S3).

2.4 | Climate Niche Representation

To determine how representative our sample is of the climate conditions experienced across the species range for each species, we created density plots of the climate values of each cell within the breeding season range raster. We then calculated what percent of the raster cells across the range has climate conditions spanned by our sample (Figure 1c). On average, our samples covered 79% of the climate conditions experienced by the species, with a range of 39%–99% and a standard deviation of 14% (Table S3, Figure S4).

2.5 | Statistical Analyses

We ran Bayesian generalized mixed models using the package **MCMCglmm** in R (Hadfield 2010). For the interspecific analysis, we used the average PC1 scores and relative masses for each of the species as the response variable. We then ran separate models for skeletal size and relative mass, with annual maximum temperature, annual minimum temperature, NDVI, and average precipitation as our predictor variables of interest, while also controlling for migratory status, habitat type, range size, and species (as a random effect). For the intraspecific analysis, we used the individual specimen PC1 and relative mass scores as the response variable, and similarly ran separate skeletal size and relative mass models in **MCMCglmm**. Our predictor variables of interest were maximum breeding temperature, minimum breeding temperature, average NDVI, average precipitation, and we additionally controlled for migratory status, habitat, range size, sex, month, and year of collection, with species as a random effect. We used uninformative inverse-Wishart priors ($V = 1$, $\nu = 0.02$) and ran the models for 55,000 iterations with a thinning rate of 1000 and burn-in of 5000. Model results for all predictor variables, including the controls, are in Table S4. We did not include latitude as a predictor because it was strongly correlated with both annual minimum temperature in the interspecific analysis ($r = -0.80$) and breeding minimum temperature in the intraspecific analysis ($r = -0.79$).

Because not all specimens had associated mass data, the intraspecific relative mass model had a smaller sample size (24 species) than the skeletal size model (57 species). To account for this, we additionally ran a skeletal size model using only the specimens included in the relative mass model.

All variables were mean-centered and converted into units of standard deviation prior to modelling. All continuous variables had variance inflation factors under 4 (Table S5), calculated

using the function *vif.MCMCglmm* (Sheard et al. 2020). We accounted for shared evolutionary history using phylogenies from BirdTree.org (Jetz et al. 2012) based on the Hackett backbone (Hackett et al. 2008). To account for uncertainty in the phylogeny, we randomly selected 100 trees from the posterior of the BirdTree dataset and pruned them to include only the species in our analysis. We then iteratively fit our model on each of the 100 trees and joined the posterior distributions, resulting in a final posterior of 5000 iterations, following (Sheard et al. 2020). We visually inspected chains to confirm proper mixing. For each model, we calculated pseudo- R^2 for both fixed effects only (mR^2) and the whole model (cR^2) following (Nakagawa and Schielzeth 2013). To further explore the sensitivity of our results to phylogenetic uncertainty, we repeated all models using a 50% majority consensus tree generated from the posterior of the BirdTree dataset, following (Rubolini et al. 2015) (Figure S5).

Pagel's λ and Blomberg's K were calculated using the *phylosig* function in the **phytools** package (Blomberg et al. 2003; Pagel and Harvey 1988; Revell 2024). To supplement these statistics, we used the 50% majority consensus tree to fit OU model parameters for skeletal size and mass, using the package **OUwie** (Beaulieu and O'Meara 2025; Cavalli-Sforza and Edwards 1967, Methods 1).

2.6 | Sensitivity Analyses

Although residuals are commonly used as a proxy for body condition (Kraft et al. 2019; Labocha and Hayes 2012; Schulte-Hostedde et al. 2005), using residuals as the dependent variable ignores the error in the estimate of the mass~PC1 regression (Freckleton 2002). To account for this, we assessed the sensitivity of the relative mass model estimates to the error in the estimates of the mass^{1/3} ~ PC1 regression. We drew 1000 samples from a normal distribution with mean and standard error corresponding to the slope estimates from the mass^{1/3} ~ PC1 regression, and 1000 samples from a normal distribution with mean and standard error corresponding to the intercept estimates. For each simulated slope and intercept estimate, we reran the model using the BirdTree consensus tree, resulting in 1000 estimates of each predictor variable. The error in the slope and intercept terms of the regression equation did not affect the direction, magnitude, or significance of the results (Figure S6).

Given that the sample size per species varied, it may be the case that species averages for PC1 and relative mass are less representative of the true mean for species where few individuals were sampled than for species where many individuals were sampled. Because at this scale across-species variation tends to be larger than within-species variation, we did not expect this to unduly influence our results. However, to ensure that differences in sample size did not affect our conclusions in the interspecific analysis, we used the consensus tree to rerun the models with each data point assigned a variance equal to the inverse of the sample size, such that points for species with greater sampling were regarded as having lower measurement error. This produced similar results to the original analysis (Table S6).

We visualized our results using the R packages **ggplot2** (Wickham 2016), **viridis** (Garnier et al. 2024), **maps** (Becker et al. 2023), **patchwork** (Pedersen 2024), **gridExtra** (Auguie 2017), **ggthemes** (Arnold 2024), and **ggdist** (Kay 2024).

3 | Results

3.1 | Interspecific Analysis

Across species, skeletal size was not associated with annual maximum temperature ($\beta = -0.03$, $pMCMC = 0.07$; Figure 2a, Table S4), annual minimum temperature ($\beta = -0.009$, $pMCMC = 0.67$), precipitation ($\beta = 0.03$, $pMCMC = 0.16$), or NDVI ($\beta = -0.02$, $pMCMC = 0.09$). In this model, the marginal R^2 was 1.6% while the conditional R^2 was 98%, indicating that patterns of phylogenetic relatedness play a dominant role in explaining variation in skeletal size across species while the environmental fixed effects explain very little.

In contrast, relative mass showed a negative relationship with both annual minimum temperature ($\beta = -0.10$, $pMCMC = 0.01$; Figure 2b, Table S4) and annual maximum temperature ($\beta = -0.07$, $pMCMC = 0.008$), such that species with ranges that are colder tended to have greater mass than expected given their skeletal size. There was no relationship between relative mass and precipitation ($\beta = 0.03$, $pMCMC = 0.36$) or NDVI ($\beta = 0.04$, $pMCMC = 0.19$). Unlike the skeletal size model, the marginal R^2 of this model was 19% while the conditional R^2 was 78% (Figure S7), suggesting that the fixed effects explained nearly 12 times more variation in relative mass than in skeletal size.

Skeletal size appears to be well-described by a Brownian motion model of trait evolution along the phylogeny (Pagel's $\lambda = 0.98$; Blomberg's $K = 0.86$, $p < 0.001$; Figure 3a), indicating that variation in overall skeletal size is statistically largely associated with evolutionary history. In contrast, relative mass shows very weak phylogenetic signal (Blomberg's $K = 0.07$, $p > 0.05$; Figure 3b) and is only moderately correlated with expectations from a Brownian motion model (Pagel's $\lambda = 0.70$). OU models further support that skeletal size evolves largely through unconstrained drift, whereas relative mass evolves under a bounded process and exhibits signs of deviation from Brownian motion consistent with adaptive convergence (Figure S8).

3.2 | Intraspecific Analysis

Within species, skeletal size was positively associated with year ($\beta = 0.02$, $pMCMC < 0.001$; Figure 2c), and males were larger than females ($\beta = 0.11$, $pMCMC < 0.001$; Table S4). Skeletal size was negatively related to maximum breeding temperature ($\beta = -0.015$, $pMCMC = 0.04$). Minimum breeding temperature, precipitation, and NDVI were not associated with skeletal size ($\beta = -0.001$, $pMCMC = 0.91$; $\beta = -0.001$, $pMCMC = 0.85$; $\beta = 0.006$, $pMCMC = 0.23$, respectively). This model explained nearly all of the variation in skeletal size (conditional $R^2 = 99\%$), but the fixed effects explained very little of the variation (marginal $R^2 = 3\%$).

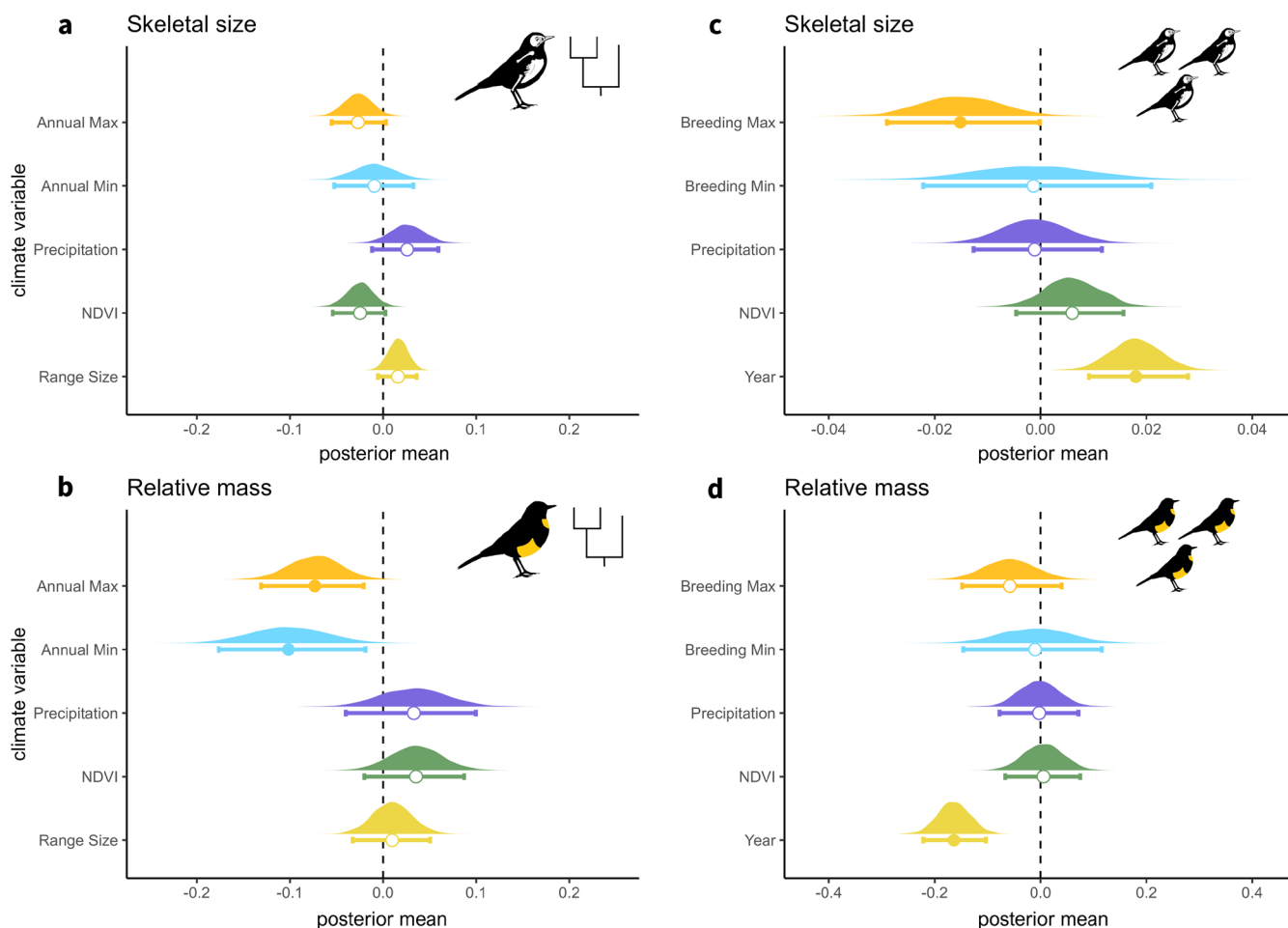


FIGURE 2 | Generalized linear mixed models results across species (a, b) and within species (c, d). Error bars display model estimates, with filled circles representing relationships where pMCMC < 0.05. Density plots show the model posterior distribution for each parameter. Bones shown in the skeletal silhouette are those used in the principal components analysis. Bird silhouettes modified from PhyloPic, courtesy of Andy Wilson (CC0 1.0).

Relative mass exhibited a negative association with year ($\beta = -0.16$, pMCMC < 0.001; Figure 2d) and was not associated with maximum breeding temperature ($\beta = -0.058$, pMCMC = 0.24), minimum breeding temperature ($\beta = 0.01$, pMCMC = 0.89), NDVI ($\beta = 0.005$, pMCMC = 0.89), or precipitation ($\beta = -0.003$, pMCMC = 0.95). Males had lower relative mass than females ($\beta = -0.27$, $p < 0.001$). The fixed effects explained 5.4% of the variation, and the full model explained 59% of the variation.

Given the different sample sizes in the intraspecific skeletal size and relative mass models, to compare the variance explained by the two models, we also fit the skeletal size model with only the specimens that were included in the relative mass model. The results were qualitatively similar to those for the larger skeletal size model, but the estimates for climate variables were no longer significant, likely due to lower power (Table S4). After normalising the sample sizes, the fixed effects explained similar amounts of variation in skeletal size and relative mass (marginal $R^2 = 5.3\%$ and 5.4% , respectively; Figure S7), but the random effect of phylogeny explained more variation in the skeletal model than in the relative mass model (conditional $R^2 = 99\%$ and 59% , respectively).

4 | Discussion

4.1 | Little Evidence of Bergmann's Rule in Passerines

We find that across a global sample of 1958 species of passerine birds, temperature does not predict interspecific variation in skeletal size. Our results are consistent with recent findings that, across species, passerine mass does not follow Bergmann's Rule in either elevational or latitudinal climatic gradients (Freeman 2017; He et al. 2023). Body size is a composite feature of allometric relationships among multiple different morphological modules. Because these modules are often highly integrated, overall body size can have limited evolvability, particularly in the short term (Araya-Ajoy et al. 2019; Orkney et al. 2021). This may be especially true for passerines. Compared to other avian taxa, passerines are characterized by relatively conserved skeletal structure (Cava et al. 2019; Navalón et al. 2022). This is reflected in the strong phylogenetic signal in skeletal size in our sample (Figure 3a), which is also apparent in other studies (Freeman 2017; He et al. 2023; Mainwaring and Street 2021). The large influence of phylogenetic inertia in determining a species' skeletal size,

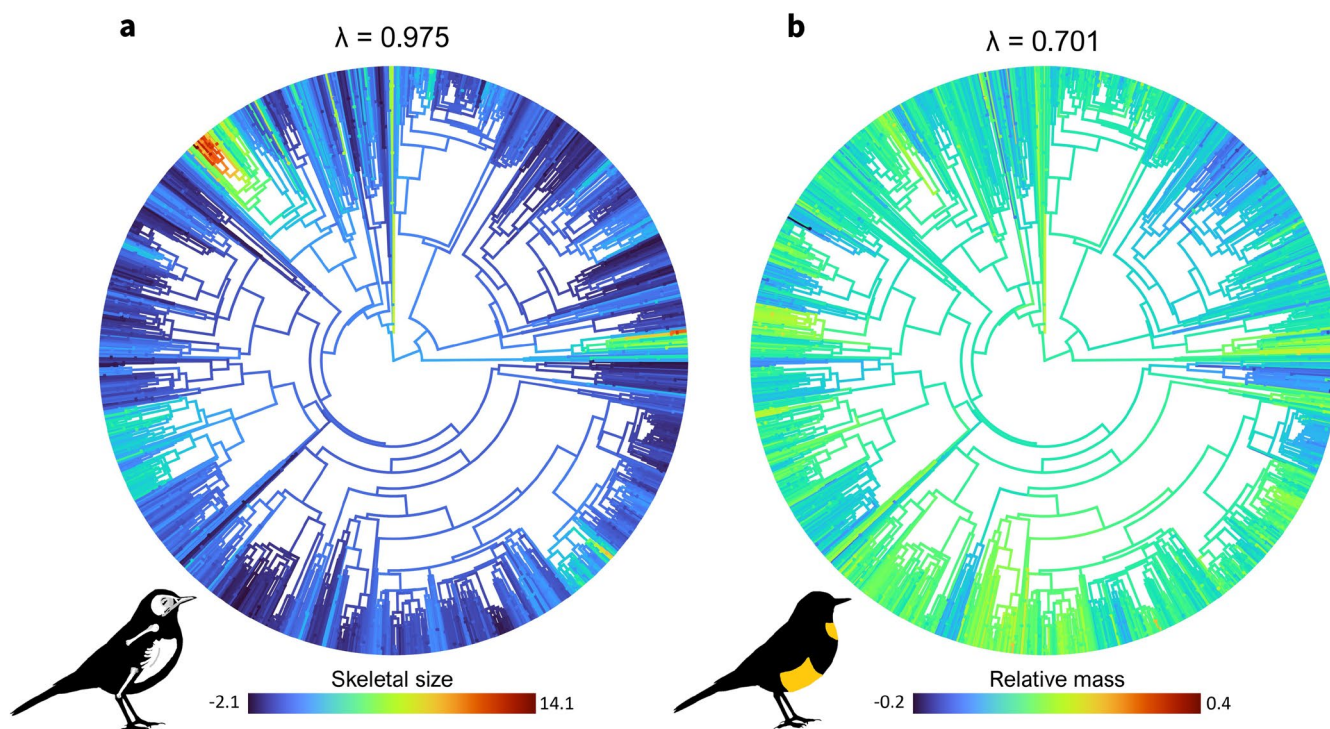


FIGURE 3 | Phylogenetic trait map of (a) skeletal size and (b) relative mass. Skeletal size exhibits stronger phylogenetic signal than relative mass. Bones shown in (a) are those used in the principal components analysis. Bird silhouettes modified from PhyloPic, courtesy of Andy Wilson (CC0 1.0).

combined with the weak support for relationships between skeletal size and environmental variables, suggests that adaptations across lineages in passerines to local climate conditions may have been primarily achieved via variation in other traits (Barve et al. 2021; Fristoe et al. 2015; Osváth et al. 2018; Ruf and Geiser 2015; Weeks, Harvey, et al. 2025).

Within 57 species, we find that temperature has a small but significant negative association with skeletal size. While our results indicate that temperature predicts some variation in size, in both the inter- and intraspecific analyses, we recover low effect sizes and find that the amount of variance in skeletal size explained by all climatic and environmental factors is small, with most variation in body size statistically explained by patterns of phylogenetic relatedness. This is broadly consistent with other recent studies that have accounted for phylogenetic relatedness in their models (He et al. 2023; Henry et al. 2023; Mainwaring and Street 2021) and suggests that environmental factors play a limited role in determining overall body size either within or across passerine species.

4.2 | Interpreting a Global Cline in Mass Relative to Skeletal Size Across Species

In contrast to body size, we find evidence of a strong association between relative mass and temperature across 1958 species of passerine birds, with species found in colder climates having larger masses than would be expected given their skeletal size. This global, interspecific cline in relative mass is consistent with ecophysiological predictions that changes in insulative materials are often more impactful for thermoregulation than equivalent changes in total body size (Briscoe et al. 2015; Steudel

et al. 1994). Furthermore, the low estimate of Blomberg's K for relative mass (see also OU model comparisons in [Supporting Information](#)) supports our hypothesis that relative mass is a trait that may correspond more to ecological conditions than evolutionary history (Blomberg et al. 2003). Combined, our results suggest that changes in insulation may be a more widespread strategy of adaptation to environmental temperature than change in overall body size.

A limitation of our study is that our relative mass index does not directly measure body composition. Thus, we cannot determine if greater relative mass represents more fat, muscle, organ, or feather mass than would be predicted from skeletal size alone. While this imprecision is a limitation of the study, feathers (Osváth et al. 2018), organs (Vézina et al. 2006), and muscle tissue (Liknes and Swanson 2011; Pani and Bal 2022) all play important roles in thermoregulation through heat generation or retention, suggesting it is reasonable to interpret relative mass in a thermoregulatory context. For instance, plumage typically comprises between 5% and 10% of a bird's body weight (Wetmore 1936), and is a bird's primary insulative structure (Herreid and Kessel 1967; Pani and Bal 2022). Species in cool environments tend to have greater feather mass than expected given their body size (Osváth et al. 2018; Wiersma et al. 2012), which may be captured in our relative mass measurement.

Past efforts have found that relative body mass is generally correlated with increased fat mass (Kraft et al. 2019; Labocha and Hayes 2012), and thus our results may be indicative of increased fat stores. Although increases in fat mass do not appear to affect thermal conductance in some species (de Vries and van Eerden 1995; Thomas et al. 1993), accumulated fat has been shown to improve heat conservation in some Arctic and

Antarctic species (Kooyman et al. 1976; Lewden et al. 2017; Mortensen and Blix 1986), as well as a widespread passerine (Blem 1974). Increased fat may also aid thermoregulation indirectly, as birds rely on fat stores for energy during periods of food scarcity which often occur in cold climates with fluctuating resources. In nine North American sparrow species, those inhabiting environments with unpredictable resources tended to have higher winter fat mass (Rogers 2015), and during a cold snap, species with increased fat deposits were more likely to survive than those with limited fat accumulation (Rogers et al. 1991). Conversely, species overwintering in the tropics tend to have lower fat deposition (Bergstrom et al. 2019). This suggests that differences in climate experienced throughout the annual cycle may drive evolved interspecific patterns in fat deposition and that these may be directly or indirectly connected to thermoregulation. Thus, regardless of the specific structures that explain the differences in relative mass, we interpret the observed cline in relative mass as evidence supporting increased insulation as a widespread adaptation that allows animals of diverse body sizes to persist in cold environments (Barve et al. 2021; Fristoe et al. 2015; Osváth et al. 2018; Scholander et al. 1950; Wiersma et al. 2012).

4.3 | Relative Mass Declines Through Time—A Signature of Global Change?

In our intraspecific analyses, we find that relative mass has declined through time but is not associated with increasing temperatures. Although testing temporal trends is not a main focus of our study, this pattern is consistent with declines in body condition that have been documented in species across the globe and, as in our study, are not linked to increases in temperature (Collins et al. 2017; Wilcox et al. 2024; but see McLean et al. 2018). Previous studies have suggested that declines in body condition may reflect habitat degradation (Wilcox et al. 2024). Declines in body condition have potentially worrying functional consequences, as birds with poor body condition have lower breeding success (Graña Grilli et al. 2018), higher mortality (Benson and Bednarz 2010), and increased vulnerability to extreme weather (Gardner et al. 2016). We emphasize that this result should be interpreted with caution, as our study was designed to test spatial, not temporal variation, and thus the rigorous repeated sampling of morphology through time that is necessary to robustly reconstruct morphological change (Weeks et al. 2026) is limited. However, our recovery of this previously-reported pattern across diverse taxa suggests that declines in relative mass through time may be a cryptic response to global change. Further study is warranted in more systems, especially in species, localities, or seasons where body condition is expected to exert significant fitness consequences.

5 | Conclusions

Climate plays an important role in shaping the body plans of animals. However, clines in overall body size along climatic gradients appear to be less general than clines in traits that have more thermoregulatory significance. We provide an example of evolved differences in one such trait: mass relative to skeletal size. Further, we find that variation in this type of trait *within*

species may be the outcome of different processes than those that gave rise to variation *among* species. As such, our results demonstrate how morphological features may operate in different contexts and be responsive to different environmental pressures at micro- versus macro-scales. Further exploration of the relationships between drivers of morphological trait variation across scales may provide insight into the mechanisms that have given rise to morphological variation within a given climate regime and the implications of this for both co-existence and adaptation to shifting environmental conditions.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in “Temperature predicts relative mass but not size in passerines” at <https://doi.org/10.5061/dryad.gf1vhhn3b>.

References

- Araya-Ajoy, Y. G., P. S. Ranke, T. Kvalnes, et al. 2019. “Characterizing Morphological (Co)variation Using Structural Equation Models: Body Size, Allometric Relationships and Evolvability in a House Sparrow Metapopulation.” *Evolution* 73, no. 3: 452–466. <https://doi.org/10.1111/evo.13668>.
- Arnold, J. B. 2024. “ggthemes: Extra Themes, Scales and Geoms for ggplot2.” <https://CRAN.R-project.org/package=ggthemes>.
- Ashton, K. G. 2002. “Patterns of Within-Species Body Size Variation of Birds: Strong Evidence for Bergmann's Rule.” *Global Ecology and Biogeography* 11, no. 6: 505–523. <https://doi.org/10.1046/j.1466-822X.2002.00313.x>.
- Ashton, K. G., M. C. Tracy, and A. de Queiroz. 2000. “Is Bergmann's Rule Valid for Mammals?” *American Naturalist* 156, no. 4: 390–415. <https://doi.org/10.1086/303400>.
- Auguie, B. 2017. “gridExtra: Miscellaneous Functions for “Grid” Graphics.” <https://CRAN.R-project.org/package=gridExtra>.
- Barve, S., V. Ramesh, T. M. Dotterer, and C. J. Dove. 2021. “Elevation and Body Size Drive Convergent Variation in Thermo-Insulative Feather Structure of Himalayan Birds.” *Ecography* 44, no. 5: 680–689. <https://doi.org/10.1111/ecog.05376>.
- Beaulieu, J. M., and B. O'Meara. 2025. “OUwie: Analysis of Evolutionary Rates in an OU Framework.” <https://github.com/thej022214/ouwie>.
- Becker, O. S., A. R. W. R. Minka, and A. Deckmyn. 2023. “Maps: Draw Geographical Maps.” <https://CRAN.R-project.org/package=maps>.

- Beltran, R. S., J. M. Burns, and G. A. Breed. 2018. "Convergence of Biannual Moulting Strategies Across Birds and Mammals." *Proceedings of the Royal Society B: Biological Sciences* 285, no. 1878: 20180318. <https://doi.org/10.1098/rspb.2018.0318>.
- Benson, T. J., and J. C. Bednarz. 2010. "Relationships Among Survival, Body Condition, and Habitat of Breeding Swainson's Warblers." *Condor* 112, no. 1: 138–148. <https://doi.org/10.1525/cond.2010.080089>.
- Bergmann, C. 1847. "Über Die Verhältnisse der Wärmeökonomie der Thiere zu über Grosso." *Gottinger Studien* 3: 595–708.
- Bergstrom, B. J., M. D. Johnson, J. C. Harris, and T. W. Sherry. 2019. "Effects of Habitat, Season, and Age on Winter Fat Storage by Migrant and Resident Birds in Jamaica." *Journal of Field Ornithology* 90, no. 2: 162–175. <https://doi.org/10.1111/jfo.12294>.
- Billerman, S. M., B. K. Keeney, P. G. Rodewald, and T. S. Schulenberg, eds. 2022. *Birds of the World*. Cornell Laboratory of Ornithology.
- BirdLife International. 2015. "BirdLife DataZone." <https://datazone.birdlife.org/contact-us/request-our-data>.
- Blem, C. R. 1974. "Geographic Variation of Thermal Conductance in the House Sparrow *Passer domesticus*." *Comparative Biochemistry and Physiology Part A: Physiology* 47, no. 1: 101–108. [https://doi.org/10.1016/0300-9629\(74\)90056-5](https://doi.org/10.1016/0300-9629(74)90056-5).
- Blem, C. R. 1976. "Patterns of Lipid Storage and Utilization in Birds." *American Zoologist* 16, no. 4: 671–684. <https://doi.org/10.1093/icb/16.4.671>.
- Blomberg, S. P., T. Garland, and A. R. Ives. 2003. "Testing for Phylogenetic Signal in Comparative Data: Behavioral Traits Are More Labile." *Evolution* 57, no. 4: 717–745. <https://doi.org/10.1111/j.0014-3820.2003.tb00285.x>.
- Briscoe, N. J., A. Krockenberger, K. A. Handasyde, and M. R. Kearney. 2015. "Bergmann Meets Scholander: Geographical Variation in Body Size and Insulation in the Koala Is Related to Climate." *Journal of Biogeography* 42, no. 4: 791–802. <https://doi.org/10.1111/jbi.12445>.
- Broggi, J., A. Gamero, E. Hohtola, M. Orell, and J.-Å. Nilsson. 2011. "Interpopulation Variation in Contour Feather Structure Is Environmentally Determined in Great Tits." *PLoS One* 6, no. 9: e24942. <https://doi.org/10.1371/journal.pone.0024942>.
- Brown, J. S., B. P. Kotler, and W. P. Porter. 2017. "How Foraging Allometries and Resource Dynamics Could Explain Bergmann's Rule and the Body-Size Diet Relationship in Mammals." *Oikos* 126, no. 2: oik.03468. <https://doi.org/10.1111/oik.03468>.
- Cava, J. A., N. G. Perlut, and S. E. Travis. 2019. "Heritability and Evolvability of Morphological Traits of Savannah Sparrows (*Passerculus sandwichensis*) Breeding in Agricultural Grasslands." *PLoS One* 14, no. 1: e0210472. <https://doi.org/10.1371/journal.pone.0210472>.
- Cavalli-Sforza, L. L., and A. W. F. Edwards. 1967. "Phylogenetic Analysis. Models and Estimation Procedures." *American Journal of Human Genetics* 19, no. 3 Pt 1: 233–257.
- Collins, M. D., G. E. Relyea, E. C. Blustein, and S. M. Badami. 2017. "Heterogeneous Changes in Avian Body Size Across and Within Species." *Journal of Ornithology* 158, no. 1: 39–52. <https://doi.org/10.1007/s10336-016-1391-x>.
- de Vries, J., and M. R. van Eerden. 1995. "Thermal Conductance in Aquatic Birds in Relation to the Degree of Water Contact, Body Mass, and Body Fat: Energetic Implications of Living in a Strong Cooling Environment." *Physiological Zoology* 68, no. 6: 1143–1163. <https://doi.org/10.1086/physzool.68.6.30163797>.
- Dufour, P., S. Descamps, S. Chantepie, et al. 2020. "Reconstructing the Geographic and Climatic Origins of Long-Distance Bird Migrations." *Journal of Biogeography* 47, no. 1: 155–166. <https://doi.org/10.1111/jbi.13700>.
- Dutcă, I., R. E. McRoberts, E. Næsset, and V. N. B. Blujdea. 2022. "Accommodating Heteroscedasticity in Allometric Biomass Models." *Forest Ecology and Management* 505: 119865. <https://doi.org/10.1016/j.foreco.2021.119865>.
- Fick, S. E., and R. J. Hijmans. 2017. "WorldClim 2: New 1-Km Spatial Resolution Climate Surfaces for Global Land Areas." *International Journal of Climatology* 37, no. 12: 4302–4315. <https://doi.org/10.1002/joc.5086>.
- Freckleton, R. P. 2002. "On the Misuse of Residuals in Ecology: Regression of Residuals vs. Multiple Regression." *Journal of Animal Ecology* 71, no. 3: 542–545.
- Freeman, B. G. 2017. "Little Evidence for Bergmann's Rule Body Size Clines in Passerines Along Tropical Elevational Gradients." *Journal of Biogeography* 44, no. 3: 502–510. <https://doi.org/10.1111/jbi.12812>.
- Fristoe, T. S., J. R. Burger, M. A. Balk, I. Khaliq, C. Hof, and J. H. Brown. 2015. "Metabolic Heat Production and Thermal Conductance Are Mass-Independent Adaptations to Thermal Environment in Birds and Mammals." *Proceedings of the National Academy of Sciences of the United States of America* 112, no. 52: 15934–15939. <https://doi.org/10.1073/pnas.1521662112>.
- Gardner, J. L., T. Amano, W. J. Sutherland, M. Clayton, and A. Peters. 2016. "Individual and Demographic Consequences of Reduced Body Condition Following Repeated Exposure to High Temperatures." *Ecology* 97, no. 3: 786–795. <https://doi.org/10.1890/15-0642.1>.
- Garnier, S., N. Ross, R. Rudis, et al. 2024. *Viridis(Lite)—Colorblind-Friendly Color Maps for R*. Zenodo. <https://doi.org/10.5281/zenodo.4679423>.
- Geist, V. 1987. "Bergmann's Rule is Invalid." *Canadian Journal of Zoology* 65, no. 4: 1035–1038. <https://doi.org/10.1139/z87-164>.
- Graña Grilli, M., M. Pari, and A. Ibañez. 2018. "Poor Body Conditions During the Breeding Period in a Seabird Population With Low Breeding Success." *Marine Biology* 165, no. 9: 142. <https://doi.org/10.1007/s00227-018-3401-4>.
- Guglielmo, C. G., Y. E. Morbey, L. V. Kennedy, J. E. Deakin, J. M. Brown, and A. T. Beauchamp. 2022. "A Scaling Approach to Understand the Dynamics of Fat and Lean Mass in Refueling Migrant Songbirds Measured by Quantitative Magnetic Resonance." *Frontiers in Ecology and Evolution* 10: 787623. <https://doi.org/10.3389/fevo.2022.787623>.
- Hackett, S. J., R. T. Kimball, S. Reddy, et al. 2008. "A Phylogenomic Study of Birds Reveals Their Evolutionary History." *Science* 320, no. 5884: 1763–1768. <https://doi.org/10.1126/science.1157704>.
- Hadfield, J. D. 2010. "MCMC Methods for Multi-Response Generalized Linear Mixed Models: The MCMCglmm R Package." *Journal of Statistical Software* 33: 1–22. <https://doi.org/10.18637/jss.v033.i02>.
- He, J., J. Tu, J. Yu, and H. Jiang. 2023. "A Global Assessment of Bergmann's Rule in Mammals and Birds." *Global Change Biology*. <https://doi.org/10.1111/gcb.16860>.
- Heldstab, S. A., C. P. van Schaik, and K. Isler. 2017. "Getting Fat or Getting Help? How Female Mammals Cope With Energetic Constraints on Reproduction." *Frontiers in Zoology* 14, no. 1: 29. <https://doi.org/10.1186/s12983-017-0214-0>.
- Henry, E., L. Santini, M. A. J. Huijbregts, and A. Benítez-López. 2023. "Unveiling the Environmental Drivers of Intraspecific Body Size Variation in Terrestrial Vertebrates." *Global Ecology and Biogeography* 32, no. 2: 267–280. <https://doi.org/10.1111/geb.13621>.
- Herreid, C. F., and B. Kessel. 1967. "Thermal Conductance in Birds and Mammals." *Comparative Biochemistry and Physiology* 21, no. 2: 405–414. [https://doi.org/10.1016/0010-406X\(67\)90802-X](https://doi.org/10.1016/0010-406X(67)90802-X).
- Hijmans, R. J. 2025. "raster: Geographic Data Analysis and Modeling." <https://rspatial.org/raster>.
- Hijmans, R. J., A. Brown, and M. Barbosa. 2026. "terra: Spatial Data Analysis [Computer Software]." <https://github.com/rspatial/terra>.

- Huston, M. A., and S. Wolverton. 2011. "Regulation of Animal Size by eNPP, Bergmann's Rule and Related Phenomena." *Ecological Monographs* 81, no. 3: 349–405. <https://doi.org/10.1890/10-1523.1>.
- Jenkins, C. N., S. L. Pimm, and L. N. Joppa. 2013. "Global Patterns of Terrestrial Vertebrate Diversity and Conservation." *Proceedings of the National Academy of Sciences of the United States of America* 110, no. 28: E2602–E2610. <https://doi.org/10.1073/pnas.1302251110>.
- Jetz, W., G. H. Thomas, J. B. Joy, K. Hartmann, and A. O. Mooers. 2012. "The Global Diversity of Birds in Space and Time." *Nature* 491, no. 7424: 444–448. <https://doi.org/10.1038/nature11631>.
- Karger, D., and N. Zimmermann. 2018. "CHELSAcruts—High Resolution Temperature and Precipitation Timeseries for the 20th Century and Beyond." *EnviDat*.
- Karger, D. N., O. Conrad, J. Böhner, et al. 2017. "Climatologies at High Resolution for the Earth's Land Surface Areas." *Scientific Data* 4, no. 1: 170122. <https://doi.org/10.1038/sdata.2017.122>.
- Kay, M. 2024. "Ggdist: Visualizations of Distributions and Uncertainty in the Grammar of Graphics." *IEEE Transactions on Visualization and Computer Graphics* 30, no. 1: 414–424. <https://doi.org/10.1109/TVCG.2023.3327195>.
- Kooyman, G. L., R. L. Gentry, W. P. Bergman, and H. T. Hammel. 1976. "Heat Loss in Penguins During Immersion and Compression." *Comparative Biochemistry and Physiology Part A, Physiology* 54, no. 1: 75–80. [https://doi.org/10.1016/S0300-9629\(76\)80074-6](https://doi.org/10.1016/S0300-9629(76)80074-6).
- Koskenpato, K., K. Ahola, T. Karstinen, and P. Karell. 2016. "Is the Denser Contour Feather Structure in Pale Grey Than in Pheomelanin Brown Tawny Owls *Strix aluco* an Adaptation to Cold Environments?" *Journal of Avian Biology* 47, no. 1: 1–6. <https://doi.org/10.1111/jav.00746>.
- Kraft, F.-L. O. H., S. C. Driscoll, K. L. Buchanan, and O. L. Crino. 2019. "Developmental Stress Reduces Body Condition Across Avian Life-History Stages: A Comparison of Quantitative Magnetic Resonance Data and Condition Indices." *General and Comparative Endocrinology* 272: 33–41. <https://doi.org/10.1016/j.ygcen.2018.11.008>.
- Labocha, M. K., and J. P. Hayes. 2012. "Morphometric Indices of Body Condition in Birds: A Review." *Journal of Ornithology* 153, no. 1: 1–22. <https://doi.org/10.1007/s10336-011-0706-1>.
- Lewden, A., M. R. Enstipp, B. Picard, T. van Walsum, and Y. Handrich. 2017. "High Peripheral Temperatures in King Penguins While Resting at Sea: Thermoregulation Versus Fat Deposition." *Journal of Experimental Biology* 220, no. 17: 3084–3094. <https://doi.org/10.1242/jeb.158980>.
- Liknes, E. T., and D. L. Swanson. 2011. "Phenotypic Flexibility of Body Composition Associated With Seasonal Acclimatization in Passerine Birds." *Journal of Thermal Biology* 36, no. 6: 363–370. <https://doi.org/10.1016/j.jtherbio.2011.06.010>.
- Mainwaring, M. C., and S. E. Street. 2021. "Conformity to Bergmann's Rule in Birds Depends on Nest Design and Migration." *Ecology and Evolution* 11, no. 19: 13118–13127. <https://doi.org/10.1002/ece3.8034>.
- McGill, B. J., B. J. Enquist, E. Weiher, and M. Westoby. 2006. "Rebuilding Community Ecology From Functional Traits." *Trends in Ecology & Evolution* 21, no. 4: 178–185. <https://doi.org/10.1016/j.tree.2006.02.002>.
- McGuire, L. P., and C. G. Guglielmo. 2010. "Quantitative Magnetic Resonance: A Rapid, Noninvasive Body Composition Analysis Technique for Live and Salvaged Bats." *Journal of Mammalogy* 91, no. 6: 1375–1380. <https://doi.org/10.1644/10-MAMM-A-051.1>.
- McLean, N., H. P. van der Jeugd, and M. van de Pol. 2018. "High Intra-Specific Variation in Avian Body Condition Responses to Climate Limits Generalisation Across Species." *PLoS One* 13, no. 2: e0192401. <https://doi.org/10.1371/journal.pone.0192401>.
- McNab, B. K. 1971. "On the Ecological Significance of Bergmann's Rule." *Ecology* 52, no. 5: 845–854. <https://doi.org/10.2307/1936032>.
- Morales-Castilla, I., M. Á. Rodríguez, and B. A. Hawkins. 2012. "Deep Phylogeny, Net Primary Productivity, and Global Body Size Gradient in Birds." *Biological Journal of the Linnean Society* 106, no. 4: 880–892. <https://doi.org/10.1111/j.1095-8312.2012.01917.x>.
- Mortensen, A., and A. S. Blix. 1986. "Seasonal Changes in Resting Metabolic Rate and Mass-Specific Conductance in Svalbard Ptarmigan, Norwegian Rock Ptarmigan and Norwegian Willow Ptarmigan." *Ornis Scandinavica* 17, no. 1: 8–13. <https://doi.org/10.2307/3676746>.
- Nakagawa, S., and H. Schielzeth. 2013. "A General and Simple Method for Obtaining R² From Generalized Linear Mixed-Effects Models." *Methods in Ecology and Evolution* 4, no. 2: 133–142. <https://doi.org/10.1111/j.2041-210x.2012.00261.x>.
- Navalón, G., A. Bjarnason, E. Griffiths, and R. B. J. Benson. 2022. "Environmental Signal in the Evolutionary Diversification of Bird Skeletons." *Nature* 611, no. 7935: 306–311. <https://doi.org/10.1038/s41586-022-05372-y>.
- Nord, A., V. Holje, B. Judik, L. P. Folkow, and P. L. Pap. 2023. "Seasonal Changes in Plumage Density, Plumage Mass, and Feather Morphology in the World's Northernmost Land Bird, the Svalbard Rock Ptarmigan (*Lagopus muta hyperborea*)." *Polar Biology* 46, no. 4: 277–290. <https://doi.org/10.1007/s00300-023-03118-8>.
- Nord, A., E. Persson, J. K. R. Tabh, and E. Thoral. 2024. "Shrinking Body Size May Not Provide Meaningful Thermoregulatory Benefits in a Warmer World." *Nature Ecology & Evolution* 8: 387–389. <https://doi.org/10.1038/s41559-023-02307-2>.
- Orkney, A., A. Bjarnason, B. C. Tronrud, and R. B. J. Benson. 2021. "Patterns of Skeletal Integration in Birds Reveal That Adaptation of Element Shapes Enables Coordinated Evolution Between Anatomical Modules." *Nature Ecology & Evolution* 5, no. 9: 1250–1258. <https://doi.org/10.1038/s41559-021-01509-w>.
- Osváth, G., T. Daubner, G. Dyke, et al. 2018. "How Feathered Are Birds? Environment Predicts Both the Mass and Density of Body Feathers." *Functional Ecology* 32, no. 3: 701–712. <https://doi.org/10.1111/1365-2435.13019>.
- Pagel, M. D., and P. H. Harvey. 1988. "Recent Developments in the Analysis of Comparative Data." *Quarterly Review of Biology* 63, no. 4: 413–440. <https://doi.org/10.1086/416027>.
- Pani, P., and N. C. Bal. 2022. "Avian Adjustments to Cold and Non-Shivering Thermogenesis: Whats, Wheres and Hows." *Biological Reviews* 97, no. 6: 2106–2126. <https://doi.org/10.1111/brv.12885>.
- Pebesma, E. 2018. "Simple Features for R: Standardized Support for Spatial Vector Data." *R Journal* 10, no. 1: 439–446. <https://doi.org/10.32614/RJ-2018-009>.
- Pebesma, E., and R. Bivand. 2023. *Spatial Data Science: With Applications in R*. Chapman and Hall/CRC. <https://doi.org/10.1201/9780429459016>.
- Pedersen, T. L. 2024. "Patchwork: The Composer of Plots [Computer Software]." <https://CRAN.R-project.org/package=patchwork>.
- Pierce, D. 2025. "ncdf4: Interface to Unidata netCDF (Version 4 or Earlier) Format Data Files [Computer Software]." <https://CRAN.R-project.org/package=ncdf4>.
- Piersma, T., and N. C. Davidson. 1991. "Confusions of Mass and Size." *Auk* 108, no. 2: 441–443.
- Pimm, S. L., C. N. Jenkins, R. Abell, et al. 2014. "The Biodiversity of Species and Their Rates of Extinction, Distribution, and Protection." *Science* 344, no. 6187: 1246752. <https://doi.org/10.1126/science.1246752>.
- Revell, L. J. 2024. "Phytools 2.0: An Updated R Ecosystem for Phylogenetic Comparative Methods (And Other Things)." *PeerJ* 12: e16505. <https://doi.org/10.7717/peerj.16505>.
- Riddell, E. A., J. L. Patton, and S. R. Beissinger. 2022. "Thermal Adaptation of Pelage in Desert Rodents Balances Cooling and

- Insulation." *Evolution* 76, no. 12: 3001–3013. <https://doi.org/10.1111/evo.14643>.
- Riener, K., R. P. Guralnick, and E. P. White. 2018. "No General Relationship Between Mass and Temperature in Endothermic Species." *eLife* 7: e27166. <https://doi.org/10.7554/eLife.27166>.
- Rising, J. D., and K. M. Somers. 1989. "The Measurement of Overall Body Size in Birds." *Auk* 106, no. 4: 666.
- Rogers, C. M. 2015. "Testing Optimal Body Mass Theory: Evidence for Cost of Fat in Wintering Birds." *Ecosphere* 6, no. 4: art55. <https://doi.org/10.1890/ES14-00317.1>.
- Rogers, C. M., J. N. M. Smith, W. M. Hochachka, et al. 1991. "Spatial Variation in Winter Survival of Song Sparrows *Melospiza melodia*." *Ornis Scandinavica* 22, no. 4: 387–395. <https://doi.org/10.2307/3676513>.
- Ross, N. 2024. "Fasterize: Fast Polygon to Raster Conversion [Computer Software]." <https://CRAN.R-project.org/package=fasterize>.
- Rubolini, D., A. Liker, L. Z. Garamszegi, A. P. Møller, and N. Saino. 2015. "Using the BirdTree.Org Website to Obtain Robust Phylogenies for Avian Comparative Studies: A Primer." *Current Zoology* 61, no. 6: 959–965. <https://doi.org/10.1093/czoolo/61.6.959>.
- Ruf, T., and F. Geiser. 2015. "Daily Torpor and Hibernation in Birds and Mammals." *Biological Reviews* 90, no. 3: 891–926. <https://doi.org/10.1111/brv.12137>.
- Schmitt, C. J., and S. V. Edwards. 2022. "Passerine Birds." *Current Biology* 32, no. 20: R1149–R1154. <https://doi.org/10.1016/j.cub.2022.08.061>.
- Scholander, P. F. 1955. "Evolution of Climatic Adaptation in Homeotherms." *Evolution* 9, no. 1: 15–26. <https://doi.org/10.2307/2405354>.
- Scholander, P. F., V. Walters, R. Hock, and L. Irving. 1950. "Body Insulation of Some Arctic and Tropical Mammals and Birds." *Biological Bulletin* 99, no. 2: 225–236. <https://doi.org/10.2307/1538740>.
- Schulte-Hostedde, A. I., B. Zinner, J. S. Millar, and G. J. Hickling. 2005. "Restitution of Mass–Size Residuals: Validating Body Condition Indices." *Ecology* 86, no. 1: 155–163. <https://doi.org/10.1890/04-0232>.
- Scott, I., P. I. Mitchell, and P. R. Evans. 1994. "Seasonal Changes in Body Mass, Body Composition and Food Requirements in Wild Migratory Birds." *Proceedings of the Nutrition Society* 53, no. 3: 521–531. <https://doi.org/10.1079/PNS19940062>.
- Sebens, K. P. 1987. "The Ecology of Indeterminate Growth in Animals." *Annual Review of Ecology and Systematics* 18: 371–407.
- Sheard, C., M. H. C. Neate-Clegg, N. Alioravainen, et al. 2020. "Ecological Drivers of Global Gradients in Avian Dispersal Inferred From Wing Morphology." *Nature Communications* 11, no. 1: 1. <https://doi.org/10.1038/s41467-020-16313-6>.
- Steudel, K., W. P. Porter, and D. Sher. 1994. "The Biophysics of Bergmann's Rule: A Comparison of the Effects of Pelage and Body Size Variation on Metabolic Rate." *Canadian Journal of Zoology* 72, no. 1: 70–77. <https://doi.org/10.1139/z94-010>.
- Stouffer, P. C., E. I. Johnson, and R. O. Bierregaard. 2013. "Breeding Seasonality in Central Amazonian Rainforest Birds." *Auk* 130, no. 3: 529–540. <https://doi.org/10.1525/auk.2013.12179>.
- Swanson, D. L. 1991. "Seasonal Adjustments in Metabolism and Insulation in the Dark-Eyed Junco." *Condor* 93, no. 3: 538–545. <https://doi.org/10.2307/1368185>.
- Tabh, J. K. R., E. Persson, M. Correia, C. Ó. Cuív, E. Thorald, and A. Nord. 2025. "Limited Evidence That Body Size Shrinking and Shape-Shifting Alleviate Thermoregulatory Pressures in a Warmer World." *Communications Biology* 8, no. 1: 1–13. <https://doi.org/10.1038/s42003-025-08131-7>.
- Tattersall, G. J., B. J. Sinclair, P. C. Withers, et al. 2012. "Coping With Thermal Challenges: Physiological Adaptations to Environmental Temperatures." In *Comprehensive Physiology*, 2151–2202. John Wiley & Sons, Ltd. <https://doi.org/10.1002/cphy.c110055>.
- Thomas, D. W., C. Bosque, and A. Arends. 1993. "Development of Thermoregulation and the Energetics of Nestling Oilbirds (*Steatornis caripensis*)." *Physiological Zoology* 66, no. 3: 322–348. <https://doi.org/10.1086/physzool.66.3.30163696>.
- Tian, L., and M. J. Benton. 2020. "Predicting Biotic Responses to Future Climate Warming With Classic Ecogeographic Rules." *Current Biology* 30, no. 13: R744–R749. <https://doi.org/10.1016/j.cub.2020.06.003>.
- Tobias, J. A., C. Sheard, A. L. Pigot, et al. 2022. "AVONET: Morphological, Ecological and Geographical Data for All Birds." *Ecology Letters* 25, no. 3: 581–597. <https://doi.org/10.1111/ele.13898>.
- Vermote, E., and NOAA CDR Program. 2019. *NOAA Climate Data Record (CDR) of AVHRR Normalized Difference Vegetation Index (NDVI), Version 5*. NOAA National Centers for Environmental Information.
- Vézina, F., K. M. Jalvingh, A. Dekinga, and T. Piersma. 2006. "Acclimation to Different Thermal Conditions in a Northerly Wintering Shorebird Is Driven by Body Mass-Related Changes in Organ Size." *Journal of Experimental Biology* 209, no. 16: 3141–3154. <https://doi.org/10.1242/jeb.02338>.
- Watt, C., and V. Salewski. 2011. "Bergmann's Rule Encompasses Mechanism: A Reply to Olalla-Tárraga." *Oikos* 120, no. 10: 1445–1447. <https://doi.org/10.1111/j.1600-0706.2011.19968.x>.
- Weeks, B. C., J. S. Berv, Y. Cho, et al. 2026. "Exploring the Data Demands and Global Opportunities for Reconstructing Morphological Responses to Climate Change." *Evolutionary Ecology* 40, no. 1: 11. <https://doi.org/10.1007/s10682-025-10368-x>.
- Weeks, B. C., C. Harvey, J. A. Tobias, C. Sheard, Z. Zhou, and D. F. Fouhey. 2025. "Longer Wing Bones in Warmer Climates Suggest a Role of Thermoregulation in Bird Wing Evolution." *Global Ecology and Biogeography* 34, no. 4: e70033. <https://doi.org/10.1111/geb.70033>.
- Weeks, B. C., Z. Zhou, B. K. O'Brien, et al. 2023. "A Deep Neural Network for High-Throughput Measurement of Functional Traits on Museum Skeletal Specimens." *Methods in Ecology and Evolution* 14, no. 2: 347–359. <https://doi.org/10.1111/2041-210X.13864>.
- Weeks, B. C., Z. Zhou, C. M. Probst, et al. 2025. "Skeletal Trait Measurements for Thousands of Bird Species." *Scientific Data* 12, no. 1: 884. <https://doi.org/10.1038/s41597-025-05234-y>.
- Wetmore, A. 1936. "The Number of Contour Feathers in Passeriform and Related Birds." *Auk* 53, no. 2: 159–169. <https://doi.org/10.2307/4077274>.
- Wickham, H. 2016. *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag.
- Wiersma, P., B. Nowak, and J. B. Williams. 2012. "Small Organ Size Contributes to the Slow Pace of Life in Tropical Birds." *Journal of Experimental Biology* 215, no. 10: 1662–1669. <https://doi.org/10.1242/jeb.065144>.
- Wilcox, R. C., T. J. Benson, J. D. Brawn, and C. E. Tarwater. 2024. "Observed Declines in Body Size Have Differential Effects on Survival and Recruitment, but no Effect on Population Growth in Tropical Birds." *Global Change Biology* 30, no. 8: e17455. <https://doi.org/10.1111/gcb.17455>.
- Zimova, M., K. Hackländer, J. M. Good, J. Melo-Ferreira, P. C. Alves, and L. S. Mills. 2018. "Function and Underlying Mechanisms of Seasonal Colour Moulting in Mammals and Birds: What Keeps Them Changing in a Warming World?" *Biological Reviews* 93, no. 3: 1478–1498. <https://doi.org/10.1111/brv.12405>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1** Interspecific consensus tree model estimates. As an additional sensitivity analysis, we re-ran our models using a 50% consensus tree. For the interspecific analysis, we also verified that transforming the PCA scores to improve normality did not affect model estimates. All analyses produced qualitatively similar results. **Figure S2.** Calculation of scaled mass residual. (A) Mass scaled with PC1 following a $1/3$ power law. (B) We regressed PC1 on $\text{mass}^{1/3}$. (C) The residuals of the $\text{mass}^{1/3} \sim \text{PC1}$ regression are heteroscedastic, with larger-bodied individuals having correspondingly larger residuals. (D) We divided each residual by the predicted value for that point. Thus, the final metric used in further analysis represents the percent deviation in mass from the predicted value. The scaled mass residual is not correlated with PC1 ($r=0.051$, D) and exhibits little to no correlation with mass ($r=0.11$, E). **Figure S3** Process models for climate data extraction. Interspecific analyses (a–d) and intraspecific analyses (e–h). Details provided in the main text, *Materials and Methods*. **Figure S4.** Intraspecific sampling for all 57 species. For the intraspecific models, sampled specimens span the breadth of the breeding range conditions. Density plots show the distribution of environmental conditions across the breeding range, with the coloured portions indicating the range of values for each variable that are covered by our sampling. Points and trendline illustrate the relationship between each environmental variable and PC1. **Figure S5.** Intraspecific consensus tree model estimates. As an additional sensitivity analysis, we re-ran our models using a 50% consensus tree. All analyses produced qualitatively similar results. **Figure S6.** Relative mass model is not sensitive to error in the equation of the $\text{mass} \sim \text{PC1}$ regression for either the interspecific (a) or intraspecific (b) model. Boxplots show the posterior means and pMCMC values for 1000 model runs with mass residuals calculated from varied regression lines. Varying the regression line did not change the significance of any variable in any run, nor did it meaningfully change the slope estimate for any significant variable. Red points indicate the values given by the model reported in the full text. **Figure S7.** Posterior distributions of marginal R^2 for interspecific (A) and intraspecific (B) skeletal size and relative mass models. Distributions were generated following Nakagawa and Schielzeth (2013). In (A) the means of the distributions significantly differ (ANOVA $p < 0.001$); in (B) the mean of the skeletal model distribution is significantly different from both the relative mass and reduced skeletal models (ANOVA $p < 0.001$, Tukey test $p < 0.001$ for both) which are marginally non-significantly different from each other (Tukey test $p < 0.054$). **Figure S8.** Disparity-through-time (DTT) plots for skeletal size and relative mass. The shaded region shows the 95% simulation envelope under a Brownian motion (BM) model of trait evolution. Skeletal size follows a trajectory that closely matches BM expectations across the phylogeny, whereas relative mass consistently exceeds BM predictions, particularly later in the tree, which indicates greater within-clade overlap and repeated evolution of similar relative mass values across subclades. **Table S1** Specimens sampled from the American Museum of Natural History. **Table S2** Loadings and variance explained for interspecific and intraspecific principal components analysis of skull, keel, tarsus, and humerus. **Table S3.** Percentage of climate niche spanned by sampled specimens. BreedingMin=minimum temperature of the breeding season; BreedingMax=maximum temperature of the breeding season; AvgPrec=average precipitation of the breeding season; Average=average of the preceding three columns. Accompanying figures for each species can be found in Figure S4. **Table S4.** Full MCMCglmm model outputs. Coefficients are followed by pMCMC in parentheses, bold indicates pMCMC < 0.05 . Intercept contains Migration: Resident, Habitat: Coastal, Sex: Female, Month: January. **Table S5** VIF scores for continuous variables used in inter- and intraspecific models. **Table S6.** Interspecific models, with species weighted by inverse of sample size. Coefficients are followed by pMCMC in parentheses, bold indicates pMCMC < 0.05 . Intercept contains Migration: Resident, Habitat: Coastal.